

Checkpoint 2:

Quantity	$k=0$ (1st step)	$k=1$ (2nd step)
$x^{(k)} = (x, y)$	$(4, 2)$	$(3.9, 1.9)$
gradient $g_k = (2x, 2y)$	$(8, 4)$	$(7.8, 3.8)$
1st moment m_{k+1}	$(0.8, 2.4)$	$(1.5, 4.44)$
2nd moment s_{k+1}	$(0.064, 0.576)$	$(0.1248, 1.0953)$
bias-corrected $\hat{m}_{k+1}$	$(8, 4)$	$(7.8947, 23.3684)$
bias-corrected $\hat{s}_{k+1}$	$(64, 576)$	$(62.4192, 547.906)$
next point $x^{(k+1)} = (x, y)$	$(3.9, 1.9)$	$(3.8001, 1.8002)$